Northern supplement to run J: 00:11 hand-off report

Core filter, early-flight family 'waypoints'. Built 19 October 2026 from the staged files on the Mac.

What was done

Run J carried few independent paths north of 30°S. Sixteen new runs put more particles in the magnetic autopilot modes. Each run also wrote a second 00:11 hand-off that takes more samples north of 34.5°S. Every sample carries an exact weight correction, so the target posterior does not change. End of flight combines these samples with run J by the balance heuristic (rule B).

Main results

1. The 00:11 latitude distribution of run J and the supplement together stays within the run-to-run spread of run J alone (Figure 1). Band masses, run J alone / combined: 30–25°S 0.0184 / 0.0194; 25–20°S 0.0023 / 0.0026.

2. The effective number of distinct 00:11 states rises from 3,987 to 13,746 in 30–25°S and from 518 to 1,049 in 25–20°S (Figure 2).

3. Moving particles between modes changes the evidence noise of each mode as 1/sqrt(particles) predicts, for all five modes within the 95 % intervals (Figure 3). The noise of the mode probabilities is therefore mainly a particle-number effect.

4. Every corrected estimate agrees with the uncorrected run: all |z| below 1.6 (Table 1). The weights add up to 1 and the program gives the same bytes as run J with the new setting off (Table 2). The 00:19 result agrees with run J (Table 3).

Limits

The 1,000-sample target applies to impact latitude under the 00:19 data options. Core can only count at 00:11; end of flight measures the target on impacts. Under rule B the northern mass rests mainly on the 16 supplement runs, so its interval does not shrink. Only the 'waypoints' family has a supplement. Run J convergence: PASS by project ruling (Pete, 16 October 2026); this report does not change it.

Contents

Figure 1 Latitude at 00:11. Figure 2 Distinct states by band and mode. Figure 3 Evidence noise by mode. Table 1 Prior recovery. Table 2 Correction checks. Table 3 Settings and results against run J and Davey et al. (2016), Fig. 10.3 (vector-recovered curve). The standard run report for the 16 runs is a separate file.

Plain: This page gives the summary of the report. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: Values quoted are those of the figures and tables that follow; sources are given in their footnotes. Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.